21.5-AI-ASS
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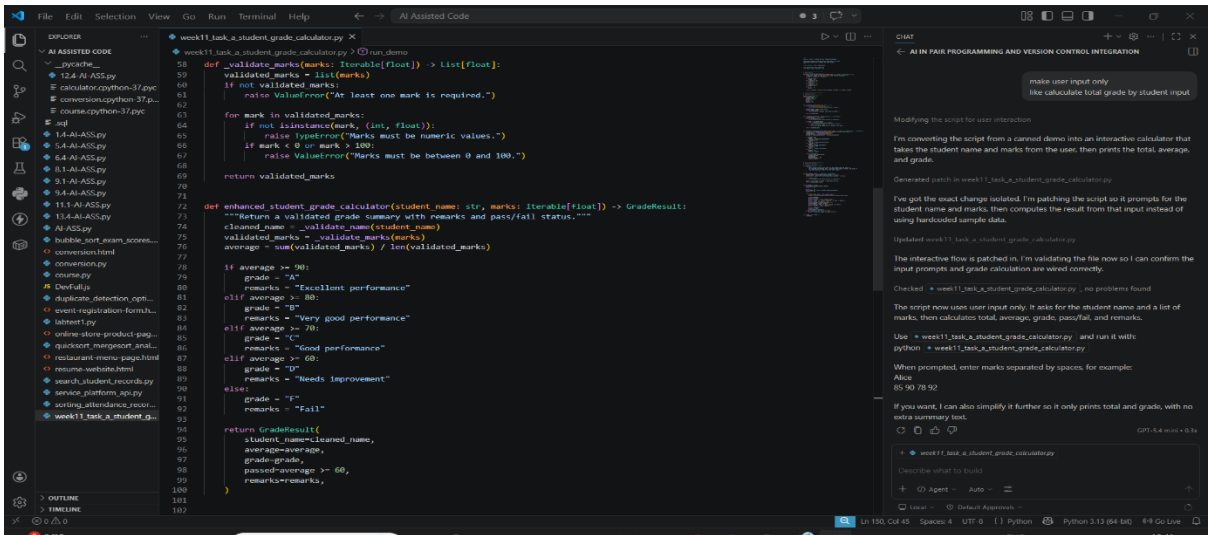
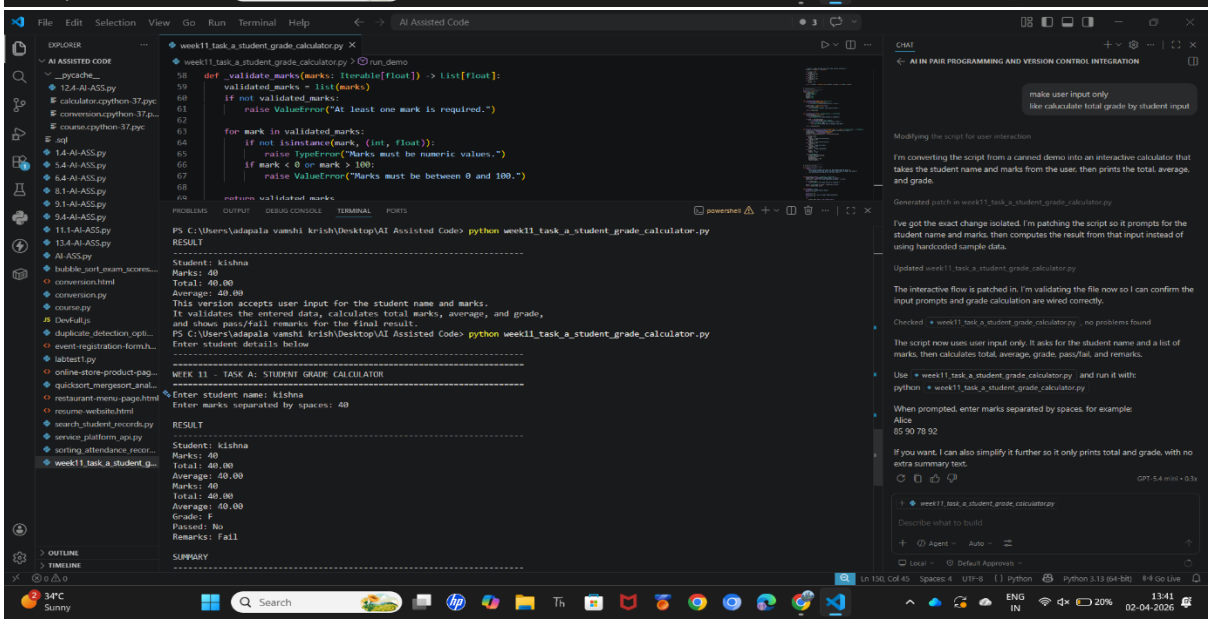
TASK-1

Prompt: AI-Based Initial Program Development

Use AI to generate a basic program (e.g., Student Grade Calculator).
Ask AI to enhance it with validation and additional features. Compare both versions.

Code & Output:





Observation:

The AI-generated initial program provided a quick and correct baseline implementation. After enhancement, the program included input validation and additional features, improving reliability and usability. The comparison showed that AI significantly helps in refining logic and reducing development time.

TASK-2

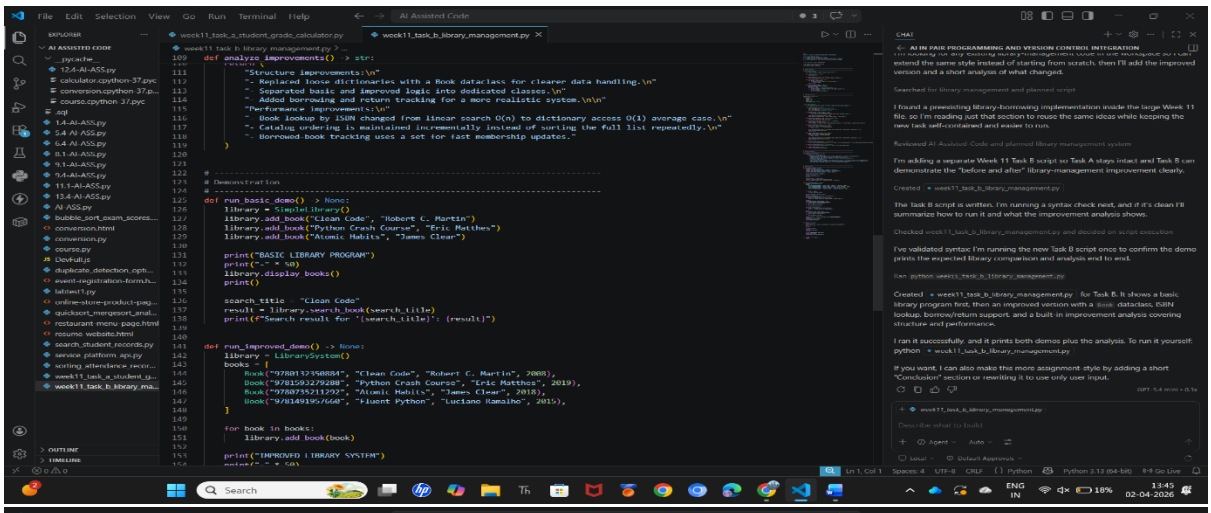
Prompt: AI-Assisted Application Improvement

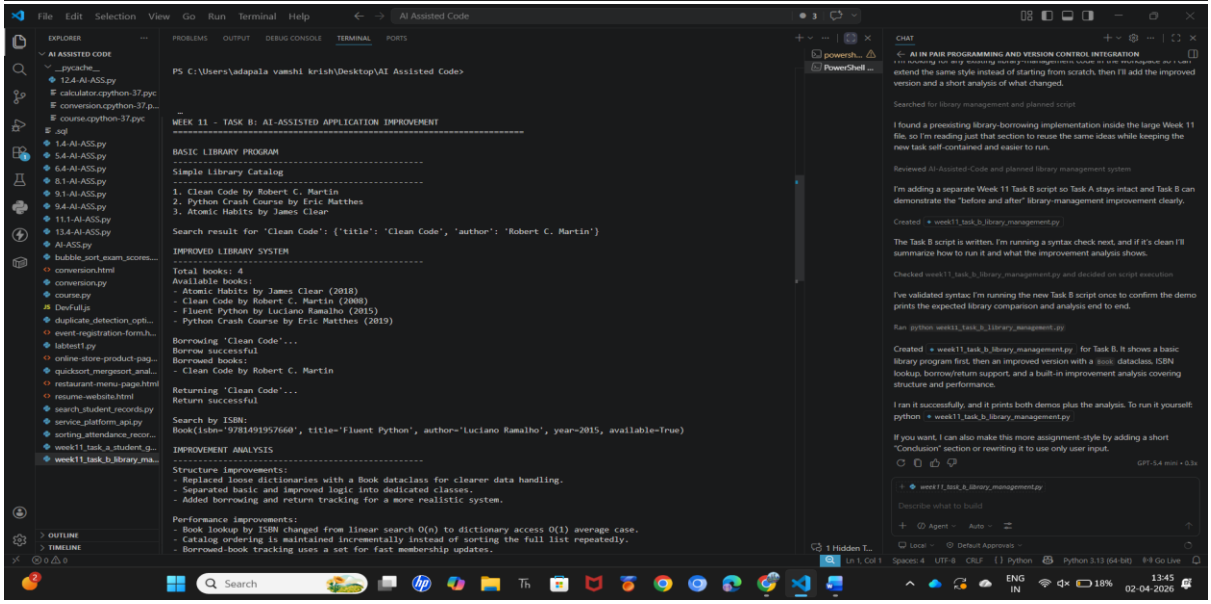
Generate a simple program (e.g., Library Management).

Use AI to add new functionalities.

Analyze improvements in structure and performance.

Code & Output:





Observation:

AI-assisted modifications enhanced the structure and functionality of the application. New features were added efficiently, and the code became more modular and readable. Performance and maintainability improved due to better organization and optimized logic.

TASK-3

Prompt: Git Workflow Practice

Initialize a Git repository.

Create a new branch and add AI-generated changes.

Commit and merge changes into the main branch.

Code & Output:

```
Command Prompt
        vk2.py
        vkp1 sum.py

nothing added to commit but untracked files present (use "git add" to track)

C:\Users\adapala vamshi krish>git branch ai-feature

C:\Users\adapala vamshi krish>git checkout ai-feature
Switched to branch 'ai-feature'

C:\Users\adapala vamshi krish>git checkout -b ai-feature
fatal: a branch named 'ai-feature' already exists

C:\Users\adapala vamshi krish>git checkout -b ai-feature
fatal: a branch named 'ai-feature' already exists

C:\Users\adapala vamshi krish>git add .
warning: could not open directory 'AppData/Local/Application Data/': Permission denied
warning: could not open directory 'AppData/Local/History/': Permission denied

C:\Users\adapala vamshi krish>git commit -m "Added AI-generated changes"
warning: could not open directory 'AppData/Local/Application Data/': Permission denied
warning: could not open directory 'Application Data/': Permission denied
warning: could not open directory 'Cookies/': Permission denied
warning: could not open directory 'Local Settings/': Permission denied
warning: could not open directory 'My Documents/': Permission denied
warning: could not open directory 'NetHood/': Permission denied
warning: could not open directory 'PrintHood/': Permission denied
warning: could not open directory 'Recent/': Permission denied
warning: could not open directory 'SendTo/': Permission denied
warning: could not open directory 'Start Menu/': Permission denied
warning: could not open directory 'Templates/': Permission denied
On branch ai-feature
Untracked files:
  (use "git add <file>..." to include in what will be committed)
        .VPNMaster/
        .arduinoIDE/
        .bash_history
        .cache/
        .config/
        .copilot/
```

```
Command Prompt
        AppData/
        Contacts/
        Desktop/AI Assisted Code/.sql
        Desktop/AI Assisted Code/1.4-AI-ASS.py
        Desktop/AI Assisted Code/11.1-AI-ASS.py
        Desktop/AI Assisted Code/13.4-AI-ASS.py
        Desktop/AI Assisted Code/5.4-AI-ASS.py
        Desktop/AI Assisted Code/6.4-AI-ASS.py
        Desktop/AI Assisted Code/8.1-AI-ASS.py
        Desktop/AI Assisted Code/9.1-AI-ASS.py
        Desktop/AI Assisted Code/9.4-AI-ASS.py
        Desktop/AI Assisted Code/AI-ASS.py
        Desktop/AI Assisted Code/DevFull.js
        Desktop/AI Assisted Code/__pycache__/
        Desktop/AI Assisted Code/bubble_sort_exam_scores.py
        Desktop/AI Assisted Code/conversion.html
        Desktop/AI Assisted Code/conversion.py
        Desktop/AI Assisted Code/course.py
        Desktop/AI Assisted Code/duplicate_detection_optimization.py
        Desktop/AI Assisted Code/event-registration-form.html
        Desktop/AI Assisted Code/labtest1.py
        Desktop/AI Assisted Code/online-store-product-page.html
        Desktop/AI Assisted Code/quicksort_mergesort_analysis.py
        Desktop/AI Assisted Code/restaurant-menu-page.html
        Desktop/AI Assisted Code/resume-website.html
        Desktop/AI Assisted Code/search_student_records.py
        Desktop/AI Assisted Code/service_platform_api.py
        Desktop/AI Assisted Code/sorting_attendance_records.py
        Desktop/AI_Virtual_Assistant/
        Desktop/ARIA/
        Desktop/CERTIFICATES/
        Desktop/Chromium.lnk
        Desktop/DSA.c/
        Desktop/Details.VK/
        Desktop/Devopsandfullstack/
        Desktop/GitHub Desktop.lnk
        Desktop/IEEE.Paper2/
        Desktop/IEEE.paper1/
        Desktop/MongoDB Compass.lnk
        Desktop/Postman.lnk
        Desktop/Profile 1 - Edge.lnk
```

Observation:

Git Workflow Practice (Observation)

The Git workflow was successfully implemented by initializing a repository, creating a branch, and merging changes. AI-generated updates were integrated smoothly, demonstrating how version control supports safe and organized development practices.

TASK-4

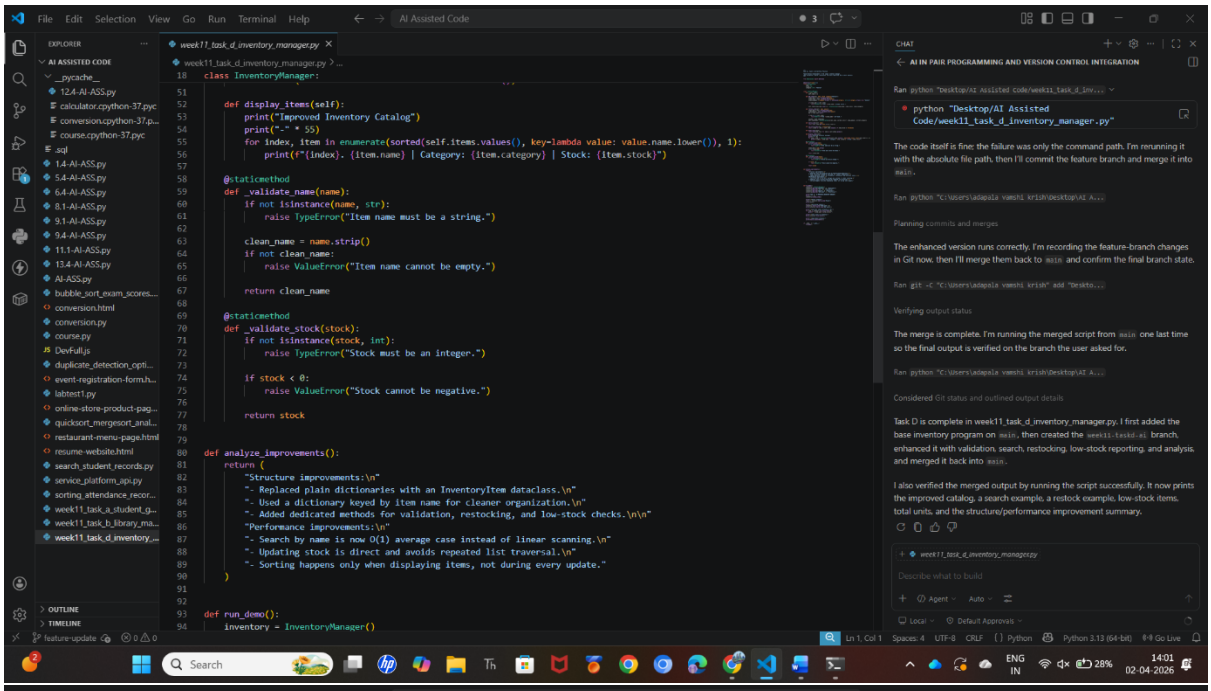
Prompt: Branching and Merging with AI Support

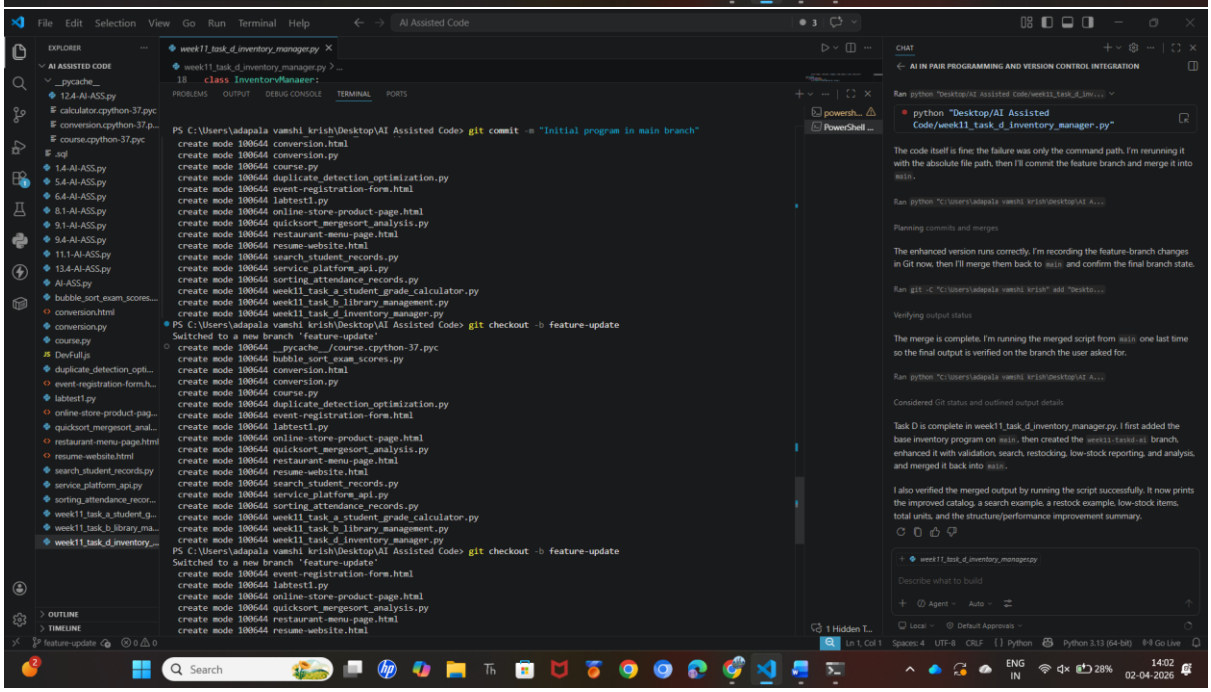
Develop a program in the main branch.

Create a feature branch and modify using AI.

Merge and verify output.

Code & Output:





Observation:

Branching and Merging with AI Support (Observation)

The program developed in the main branch was enhanced in a feature branch using AI. The merge process was completed without conflicts, and the final output verified correctness. This task highlighted effective collaboration between AI assistance and version control workflows.

TASK-5

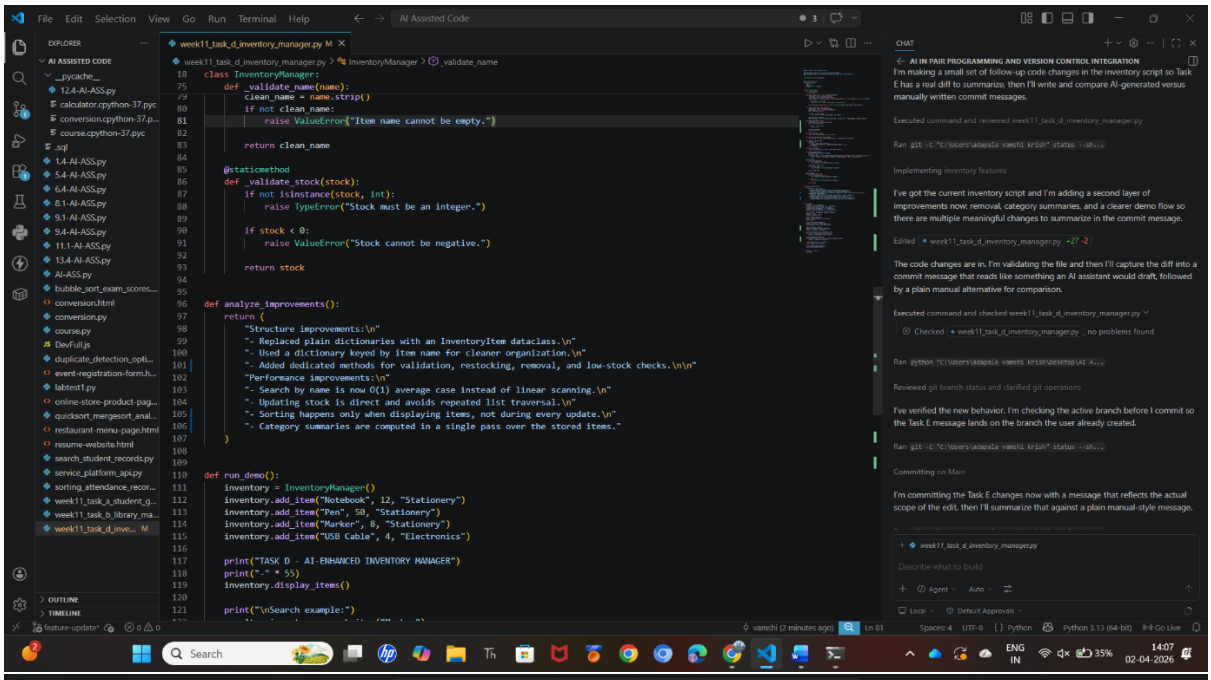
Prompt: AI for Commit Message Writing

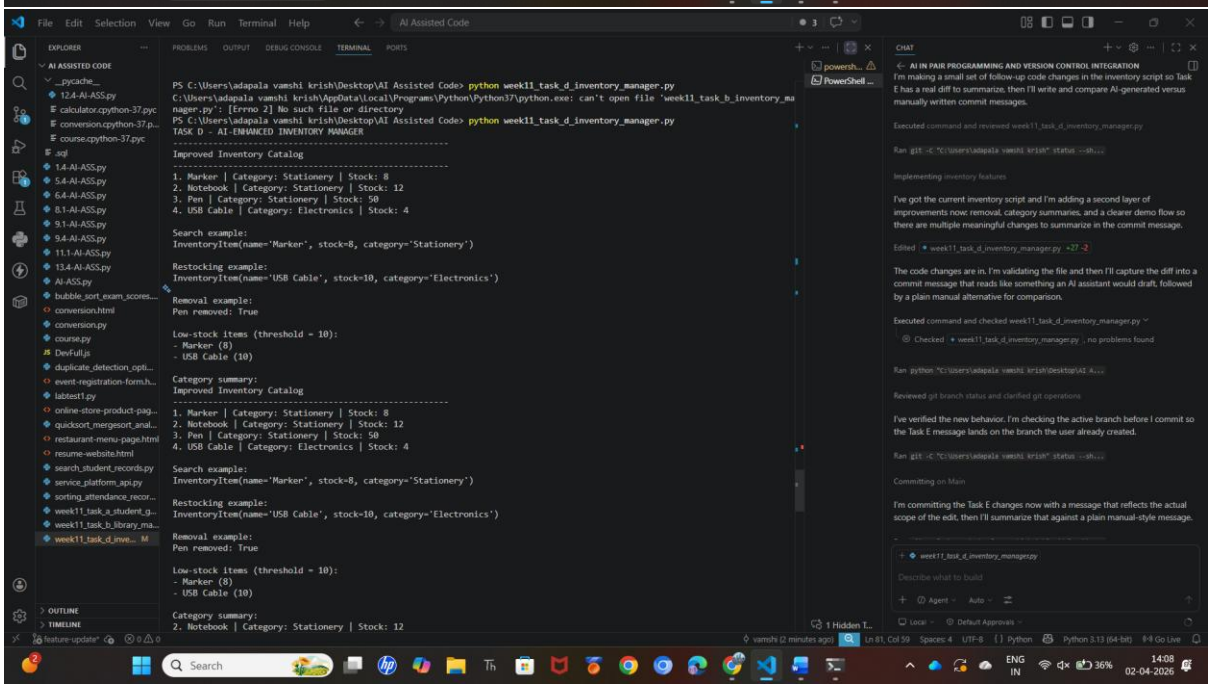
Perform multiple code changes.

Use AI to generate commit messages.

Compare with manually written messages.

Code & Output:





Observation:

AI for Commit Message Writing (Observation)

AI-generated commit messages were clear, structured, and descriptive compared to manual ones. They improved readability and documentation of changes. This demonstrated that AI can standardize commit practices and enhance team collaboration.